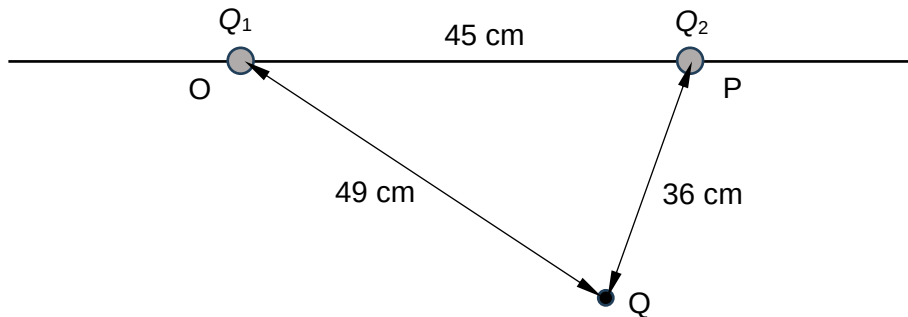


- 18** Two fixed point charges, each of magnitude $Q_1 = -7.4 \text{ nC}$ and $Q_2 = +3.6 \text{ nC}$, are located at O and P respectively as shown. The distance OP is 45 cm.



The distances between O, P and Q are shown in the figure.

What is the value of the potential at point Q?

- A** -120 V **B** -56 V **C** -46 V **D** 220 V

Ans: C

$$V = V_1 + V_2 = \frac{Q_1}{4\pi\epsilon_0 r_1} + \frac{Q_2}{4\pi\epsilon_0 r_2} = \frac{1}{4\pi\epsilon_0} \left(\frac{Q_1}{r_1} + \frac{Q_2}{r_2} \right) = -46 \text{ V}$$

- 19** Which one of the following statements about the electric potential and electric field at a point is